Eason Huang

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Education

University of Waterloo — Waterloo, ON
Bachelor of Computer Science — *Expected 2030*

- **Schulich Leader Scholarship (UofT)**: $120k engineering award, selected nationally; declined to attend Waterloo
- **Ronald G. Scoins National Scholarship**: $18k entrance scholarship for outstanding students

Technical Skills

Languages: C++, C, Python, CUDA, Rust, SQL, Java, Go, TypeScript, HTML/CSS
Frameworks & Tools: PyTorch, TensorFlow, Triton, LangChain, Docker, Kubernetes, PostgreSQL, Linux, Git, Bash

Experience

WAT.ai (Waterloo AI Design Team) — Sept 2026 – Present
Research Engineer, Compute Team — *Waterloo, ON*

- Building GPU inference kernels for an adaptive TTS engine that predicts per-frame refinement steps.
- Optimizing PyTorch and Triton kernels, profiling to turn reduced compute into real wall-clock speedup.
- Benchmarking against existing step-reduction methods, targeting an open-source release and paper submission.

Nokia (Cloud & Network Services) — May 2025 – Aug 2025
AI/ML Developer — *Kanata, Ontario*

- Built an MCP/A2A agent on Kubernetes that learns from 5G technical docs and keeps long-term memory.
- Combined BM25, vector search and a Neo4j knowledge graph to search 3000+ pages of network documentation.
- Built structure-aware YAML/XML/YANG chunking that keeps parent key paths, improving recall@10 by 26%.
- Made network diagrams searchable by text using CLIP embeddings and OCR.
- Fine-tuned Qwen-7B with QLoRA and GRPO on TeleQnA to specialize it in 5G and 3GPP standards.

Factful (Startup) — July 2023 – Nov 2025
Co-founder — *Remote*

- Developed a RAG-based meta-search engine, implemented semantic reranking that filters non-authoritative sources, raising Precision@5 by 25%.
- Reverse-engineered Google Docs' internal API to power a Chrome extension
- Migrated Azure App Service to AKS, cutting cloud costs 40% and enabling autoscaling.
- Led a team of 5 to ship the platform (Flask, React, Azure), scaling to 20k+ visits in beta.

Projects

CPU Matmul from Scratch | *C++, perf*

- Wrote a matrix-multiply (GEMM) kernel for Zen 5 CPUs that runs 3% faster than OpenBLAS.
- Implemented an AVX-512 microkernel with C++ intrinsics, hierarchical tiling, memory packing and autotuning.
- Profiled IPC and L3 cache misses with perf and modeled memory bandwidth to guide tiling decisions.

GPU Matmul from Scratch | *CUDA, C++, ncu*

- Wrote a ladder of CUDA GEMM kernels from naive to warp-tiled, vectorized and register-tiled.
- Tuned the warp-tiled kernel to 8.7 TFLOP/s on an RTX 3060, beating cuBLAS by 12% on FP32 GEMM.
- Profiled with Nsight Compute to find stalls, and autotuned tile configs over 8 parameters.

Bricked (Hack the North 2026 Winner) | *Python, Next.js, Three.js, Claude API*

- Built a multi-agent pipeline (planner, builder, inspector) that turns a sentence into a buildable 3D LEGO model.
- Wrote a stability checker using force and torque balance to reject models with floating or toppling bricks.
- Streamed each build step to a Next.js and Three.js viewer over Server-Sent Events.

Neural Network From Scratch | *Python, NumPy*

- Built an MLP in pure NumPy with hand-derived backpropagation and gradient descent, reaching 98% on MNIST.
- Replaced per-sample Python loops with batched matrix operations, speeding up training by over 500x.